

Chapter 7: Independence of the Parallel Postulate:

P-exercises:

P-1. Translations between Poincare and Euclidean definitions:

a) Poincare: IF two triangles are equivalent in sides and angles, then they are congruent.

Euclidean: IF two triangles are equivalent in sides and angles, then they are congruent.

b) Poincare: IF two lines are divergently parallel, then they have a common perpendicular and the latter is unique.

Euclidean: IF two lines are divergently parallel, then they have a common perpendicular.

P-2. (from P-exercise) "When O lies inside δ

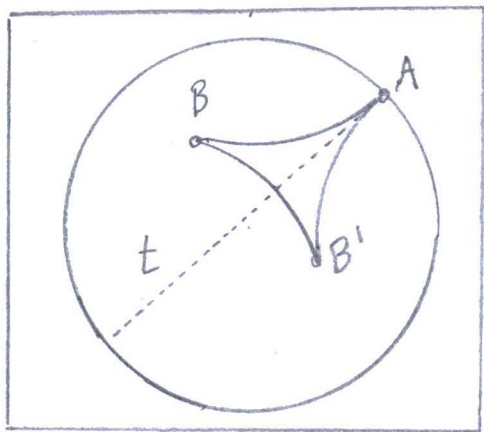
and the power p of O with respect to δ is negative."

(pg 322) "Analogue to Proposition 7.6"

"Let γ be a circle of radius r and center O , δ a circle of radius s and center C . Assume O lies inside δ ; let p be the power of O with respect to δ . Let $K = r^2/p$. Then the image C^* of C under the dilation from O of ratio K . If P is any point on δ and P' is its inverse in γ , then the tangent t' to δ' at P' is the reflection across the perpendicular bisector of PP' of the tangent to δ at P .

P-3. (from P-exercise) " A' is the inverse of C
in the image of α under
the inversion of $\mathcal{J}$ "

P-4



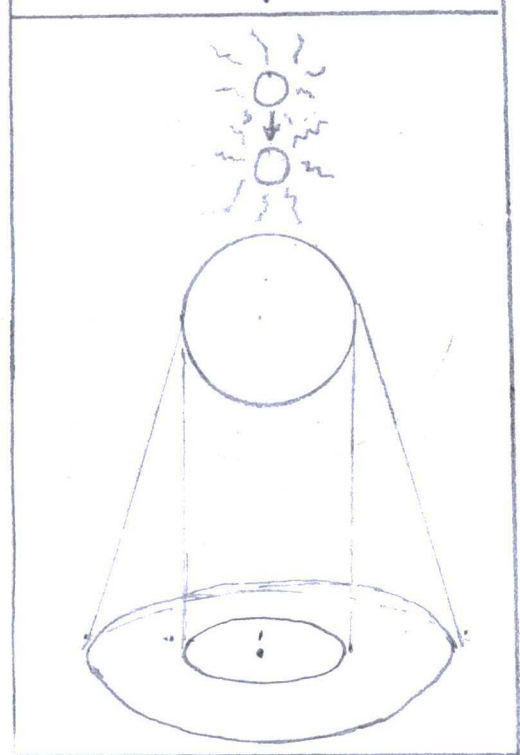
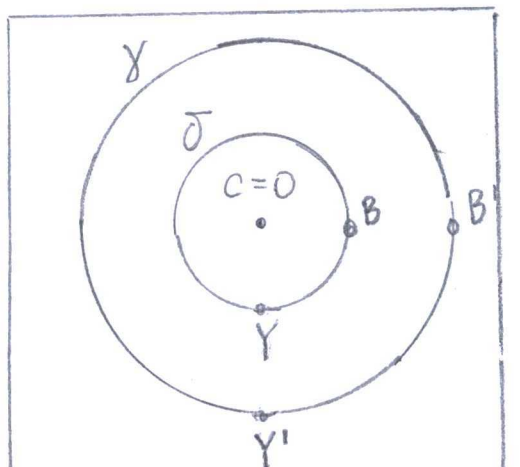
"P-angle-bisector"

Line $t = P$ -angle bisector

$\angle BAB' = P$ -angle

P-5

Construction of a P-center from an E-circle:



"P-center ($C=O$)"

case 1 ($G=0$):

- 1) The circle δ has center C .
- 2) Point C equals point O to another larger circle γ .
- 3) (E)-circle δ maps onto (P)-circle γ by distance (Lemma 7.4), and Axiom C-6(SAS). The P-center is also the E-center $[C]$. Points Y, Y', B, B' represent before and after transformation.

Case 2 ($C \neq O$):

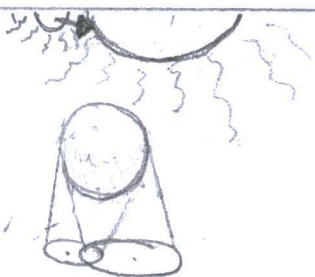
1) Circle γ has radius OA with point B on the perimeter.

2) $\overrightarrow{OA}$ traverses point C .

3) CB is circle δ

4) (E)-circle γ maps to (P)-circle δ by distance (Lemma 7.4), and $S.A.S.$

"P-center ($C \neq O$)"



Case 3 ($C \neq O, A \neq B \neq O$):

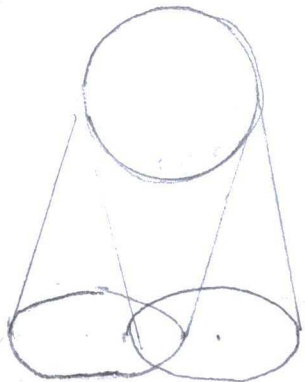
1) Circle γ has radius OA

2) $A \neq B \neq O$

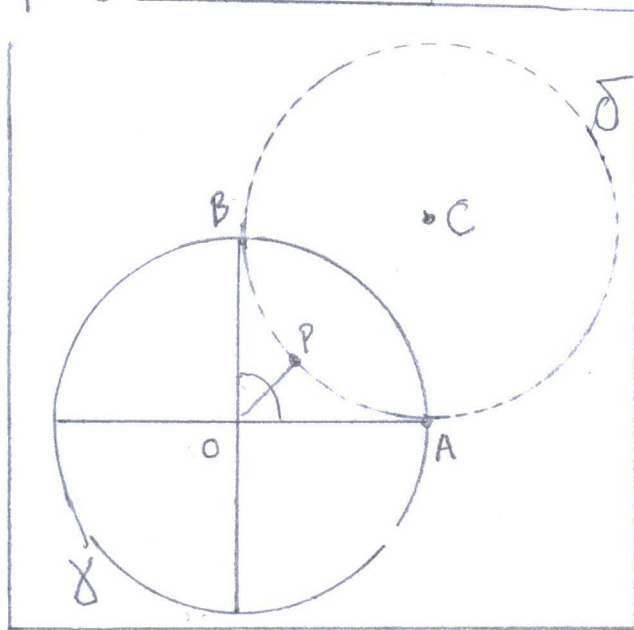
3) Point C is OA away from B

4) Circle δ has radius OA ,

"P-center ($C \neq O, A \neq B \neq O$)"



P-6



"Schweikart's constant"

Schweikart's constant by
Cross ratio:

(pg 319) "Cross ratio"

$$\begin{aligned} (OP, AB) &= \frac{\overline{OB}}{\overline{OA}} \cdot \frac{\overline{PB}}{\overline{PA}} \\ &= \frac{\overline{PB}}{\overline{PA}} \\ &= \frac{r + \overline{OP}}{r - \overline{OP}} \end{aligned}$$

(pg 320) "Poincare length"

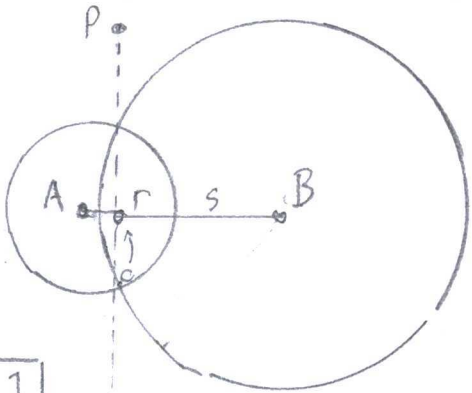
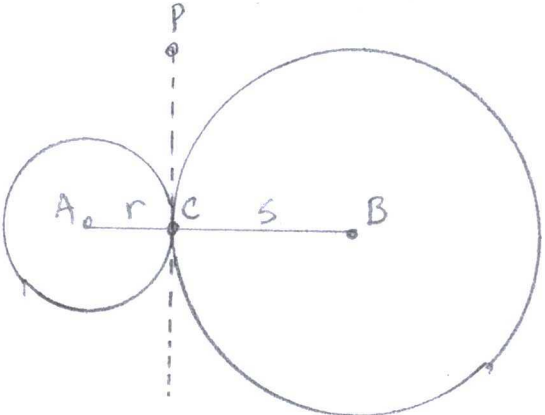
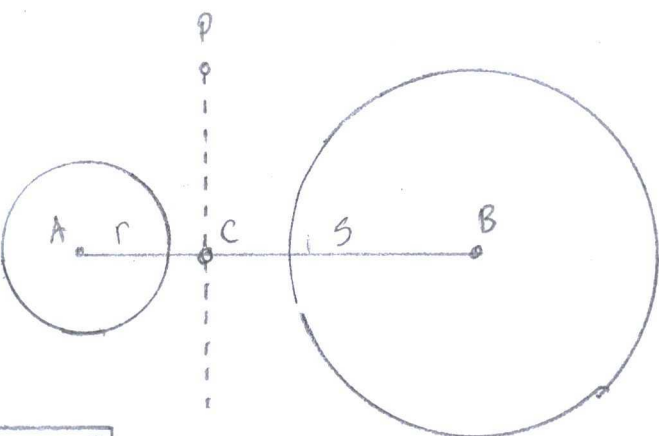
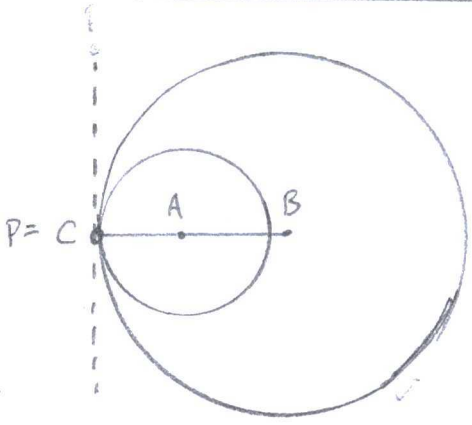
$$d(OP) = |\log(OP, AB)|$$

If $r=1$, then $e^d = \frac{1 + \overline{OP}}{1 - \overline{OP}}$ and

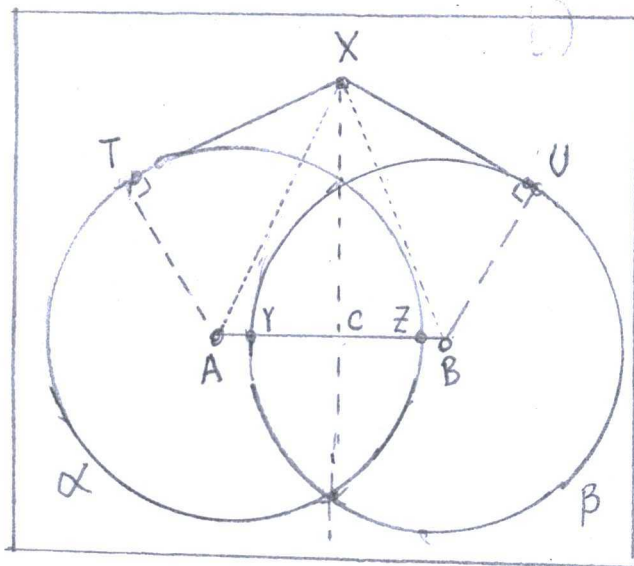
the Euclidean distance of $\overline{OP} = \sqrt{2} - 1$
is $d = \log(\sqrt{2} - 1)$.

P-7

a) (from P-exercise) "C exists and is
unique and that for any point
P different from A and B, P lies
on the radical axis if and only
if $(\overline{PA})^2 - (\overline{PB})^2 = r^2 - s^2$

Proof: Hypothesis	Evidence
Case 1 	Case 1-4 1) Circle α has radius r from center A Euclid's Postulate III
Case 2 	2) Circle β has radius s from center B Euclid's Postulate III 3) $A \neq B$ (pg 14) "Definition-point"
Case 3 	4) Point C lies on $\overline{AB}$ such $\overline{AC}^2 - \overline{BC}^2 = r^2 - s^2$ (pg 14) "Definition-lies on"
Case 4 	5) A line through C is perpendicular to AB (pg 359) "Definition-Radical axis" 6) Point P appears on the radical axis, so $\overline{PA}^2 - \overline{PB}^2 = r^2 - s^2$. Because P has exact distance to external tangents Pythagoreas Theorem

*Note $\overline{AC}$ accurate to $\pm 10\%$ $\overline{AB}$.



"radical axis"

- 1) Circle α has radius r from center A .
- 2) Circle β has radius s from center B .
- 3) $A \neq B$
- 4) Point C lies on AB
so $AC^2 - BC^2 = r^2 - s^2$
- 5) A line through C is perpendicular to AB .
- 6) Point X lies on the radical axis

b) (from P-exercise) " $\overline{XT} = \overline{XU}$ if and only if X lies on the radical axis of α and β ."

Proof: Hypothesis | Evidence

Euclid's Postulate III

Euclid's Postulate III

(pg 14) "Definition-point"

(pg 14) "Definition-lies on"

(pg 35a) "Definition-Radical axis"

RAA hypothesis

of α and β .

7) $\overline{XT}$ is tangent to α at T

8) $\overline{XU}$ is tangent to β at T

9) $\overline{XY} = \overline{XZ} = \overline{XU} = \overline{XT}$ at the radical axis

$$10) AT^2 + \overline{XT}^2 = \overline{AX}^2$$

$$\text{and } BU^2 + \overline{XU}^2 = \overline{BX}^2$$

$$\overline{AX}^2 - \overline{BX}^2 = AT^2 - BU^2 = r^2 - s^2$$

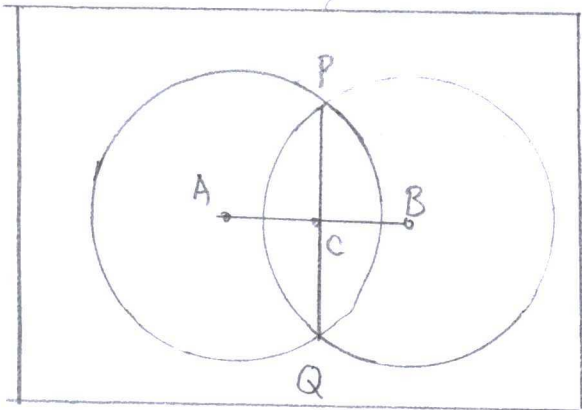
(pg 202) "Definition-tangent"

(pg 202) "Definition-tangent"

Axiom C-2

RAA result

c)



(from P-exercise) "If α and β intersect in two points

P and Q , PQ is the radical axis."

Proof: Hypothesis

Evidence

1) Circle α has radius r from center A | Euclid's Postulate III

2) Circle β has a radius s from center B . | Postulate III

3) $A \neq B$

4) Point C lies ...

... on AB, so

$$AC^2 - BC^2 = r^2 - s^2$$

5) A line through C is perpendicular to AB

6) Circles α and β intersect at P and Q

7) Points P and Q are on the radical axis because $AP^2 = CP^2 + AC^2$ and $BP^2 = CP^2 + BC^2$ for $r^2 - s^2 = AC^2 - BC^2$

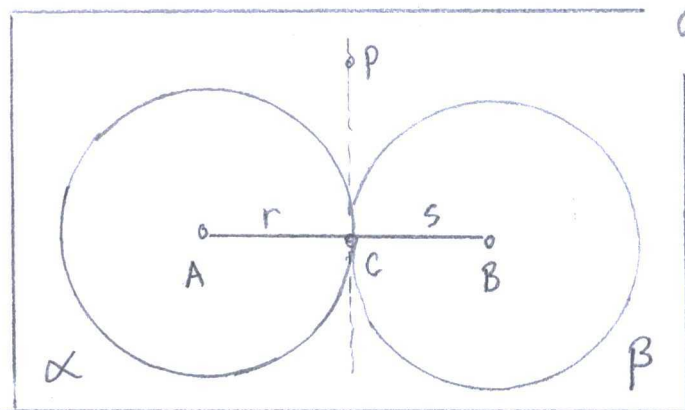
(pg 14) "Definition - Point"

(pg 14) "Definition - lies on"

(pg 359) "Definition - Radical axis"

Circle-Circle Continuity Principle

RAA result



d) (from P-exercise) "If α and β are tangent at point C, the radical axis is the common line through C"

Proof Hypothesis	Evidence
1) Circle α has a	Euclid's Postulate III

radius r from center A .

2) Circle β has radius s from center B

3) $A \neq B$

4) Point C lies...
... on AB .

$$\text{so } AC^2 - BC^2 = r^2 - s^2$$

5) A line through C is perpendicular to AB .

6) Point P is on the radical axis, so

$$PA^2 = AC^2 + CP^2,$$

$$PB^2 = BC^2 + CP^2, \text{ and}$$

$$PA^2 - PB^2 = r^2 - s^2.$$

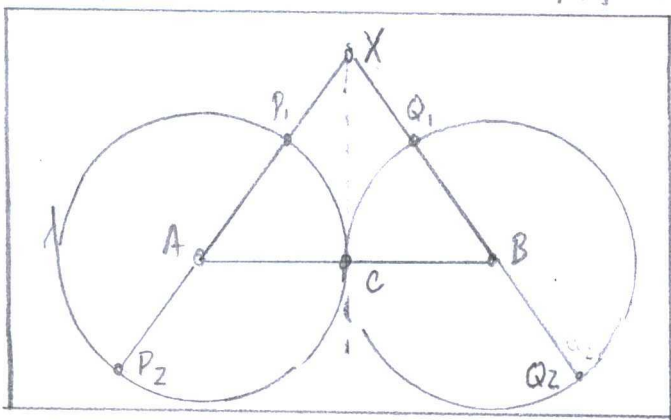
Euclid's
Postulate III

(pg 14) "Definition-
Point"

(pg 14) "Definition-
lies on"

(pg 359) "Definition
Radical axis"

RAA result



e) (From P-exercise) "X lies on the radical axis of α and β if and only if X has the same power with respect to α and β "

Proof: Hypothesis

Evidence

1) Circle α has radius r and center A

Euclid's
Postulate III

2) Circle β has radius s and center B .

Euclid's
Postulate III

3) $A \neq B$

(pg 14) "Definition - point"

4) Point C lies $\dots$
 $\dots$ on AB , so

(pg 14) "Definition -
lies on"

$$AC^2 - BC^2 = r^2 - s^2$$

5) A line through C is perpendicular to AB .

(pg 359) "Definition -
Radical axis"

6) Point X is on the radical axis because the same power to α and β .

Lemma 7.1

$$XC^2 = \overline{XA}^2 - AC^2$$

$$= (\overline{XA} - AC)(\overline{XA} + AC)$$

$$= (\overline{XA} - AP_1)(\overline{XA} + AP_2)$$

$$= (\overline{XP_1})(\overline{XP_2})$$

$$XC^2 = XB^2 - BC^2$$

$$= (\overline{XB} - BC)(\overline{XB} + BC)$$

$$= (\overline{XB} - BQ_1)(\overline{XB} + BQ_2)$$

$$= (\overline{XQ_1})(\overline{XQ_2})$$

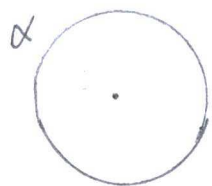
$$XP_1 \cdot XP_2 = XQ_1 \cdot XQ_2$$

P-8

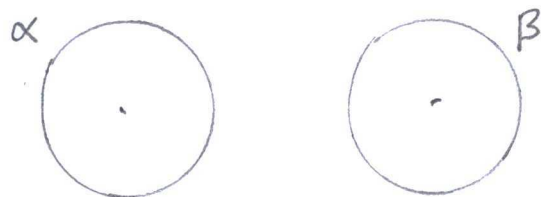
(From P-exercise) "Circle σ that cuts α in two points A' and A'' and β in two points B' and B'' ."

By straightedge and compass:

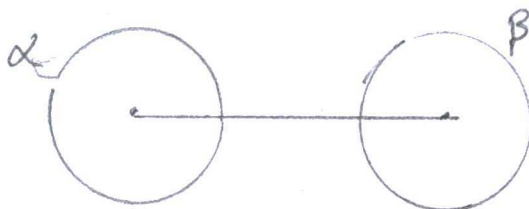
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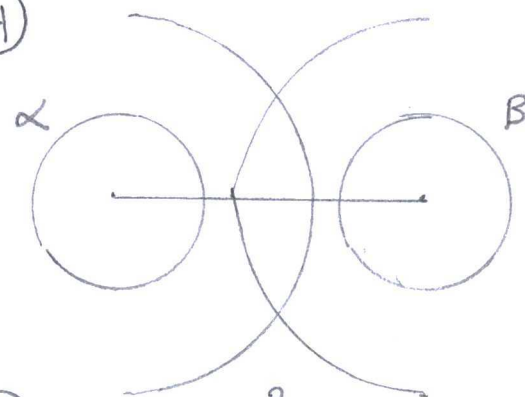
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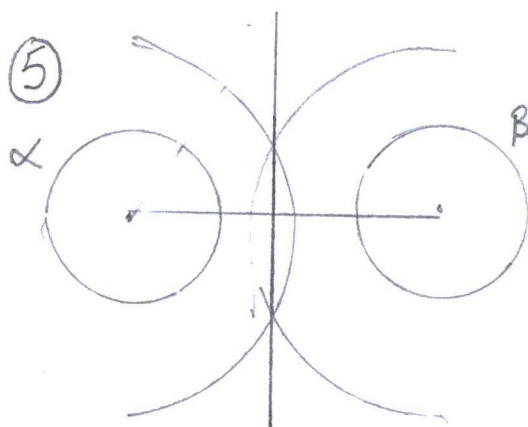
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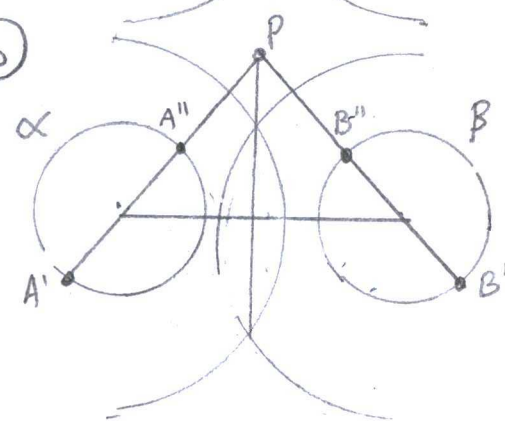
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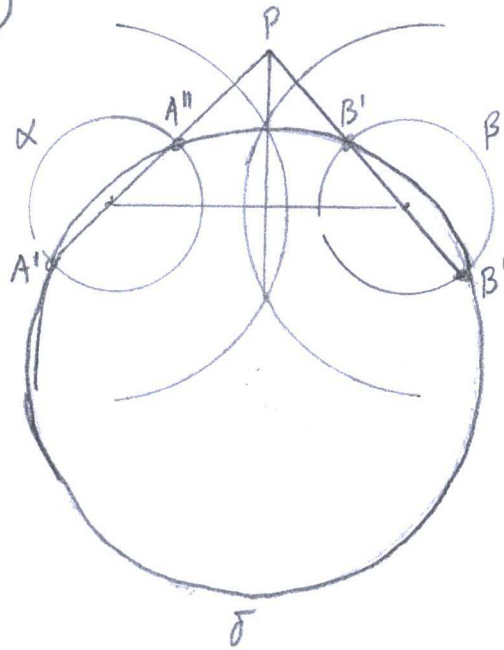
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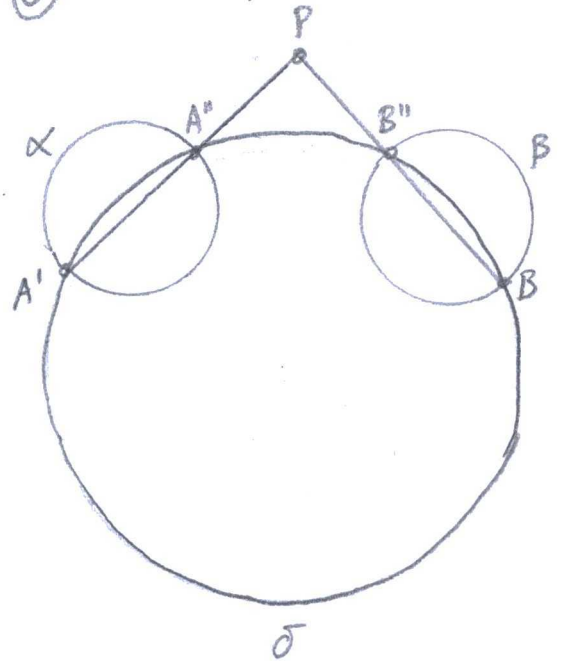
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⑦



⑧



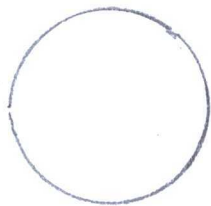
P-9.

(from P-exercise) "In the Poincaré model, two divergently parallel lines have a common perpendicular."

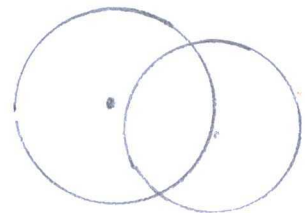
By straightedge and compass:

Case 1: $(A \parallel B) \perp C$

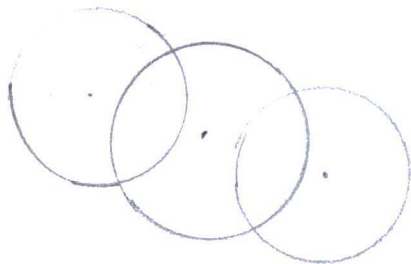
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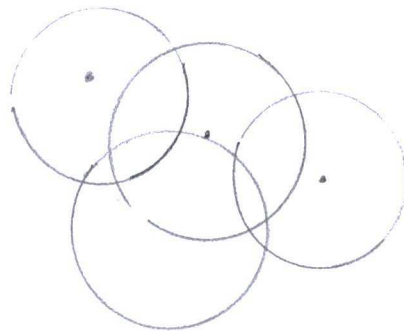
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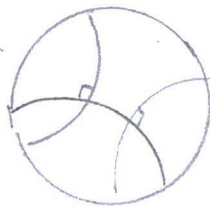
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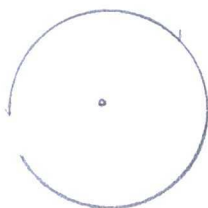


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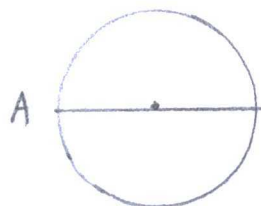


Case 2: $(A \parallel B) \perp C$ & $A \perp O$.

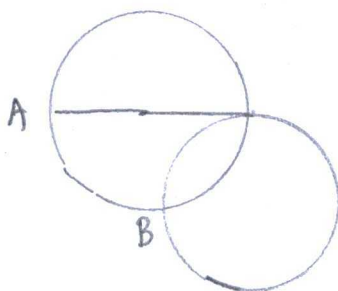
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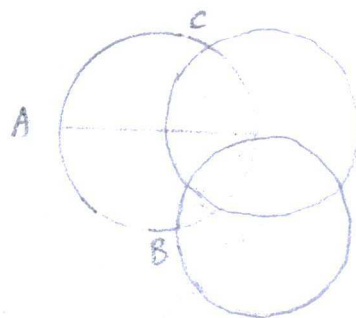
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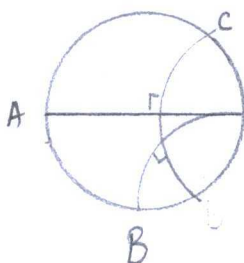
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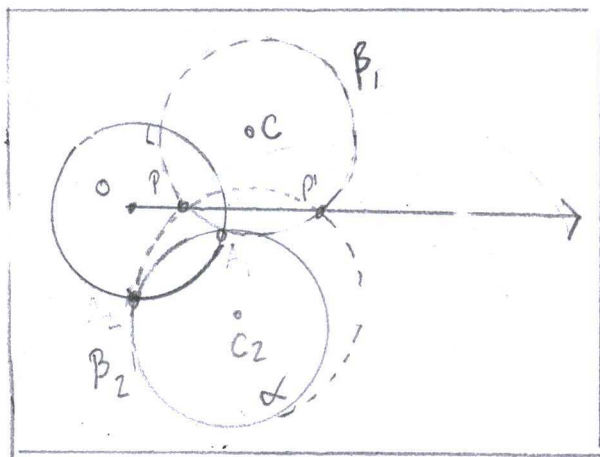
④



⑤



P-10



"Poincaré limiting
parallel rays"

Two rays from P in the Poincaré
model that are limiting parallel to l:

β_1 : From center O, a ray $\overrightarrow{OP}$

An inverse point P' is necessary
on ray OP by Proposition 7.5,
such $(OA)^2 = (OP)(OP')$

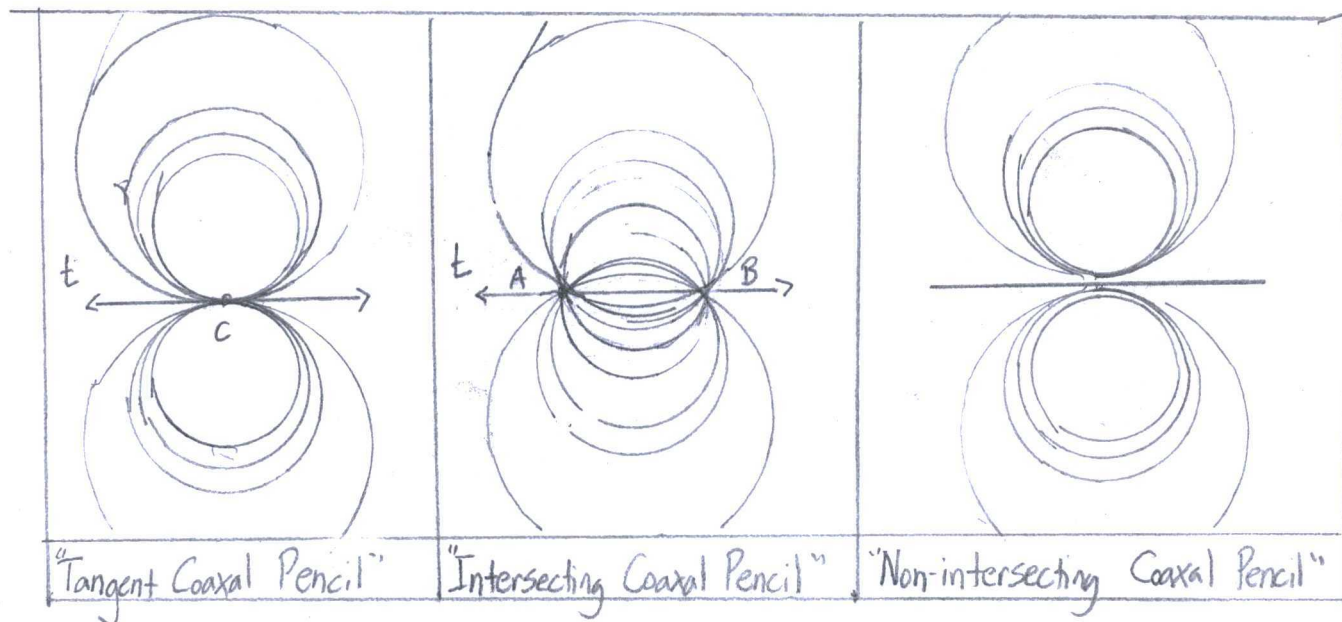
A center C exists perpendicular
to PP'

The center to β_1 is by C,

With radius $CP = C_1P'$

β_2 : A center C_2 defines circle β_2
from a perpendicular upon $\overrightarrow{OP}$.

β_2 has radius $CP = C_2P'$



"Tangent Coaxal Pencil"

"Intersecting Coaxal Pencil"

"Non-intersecting Coaxal Pencil"

(from P-exercise) "Any two nonconcentric circles belong to a unique coaxal pencil"

Proof: Hypothesis

Evidence

Case 1: Circles intersect at two points

A) Circles α and β intersect at two points A and B.

B) Segment AB is a line $\vec{l}$ in type II coaxal pencils

Circle-Circle continuity Principle

(pg 361) "Definition-Intersecting Coaxal Pencil"

Case 2: Circles intersect at one point

A) Circles α and β intersect at one point C

B) Line $\vec{l}$ is perpendicular to OO'

Circle-Circle Continuity Principle

Proposition 4.1

C) α, β , and ℓ are a tangent coaxal pencil

(pg 361) "Definition-tangent coaxal pencil"

Case 3: Circles intersect at no point

A) Circles α and β intersect at no point

B) A line $\overleftrightarrow{OO'}$ has a midpoint C .

C) Line ℓ is perpendicular to OO' at C

D) Line ℓ is a radical axis

E) The conditions are non-intersecting coaxal pencils

Circle-Circle Continuity Principle

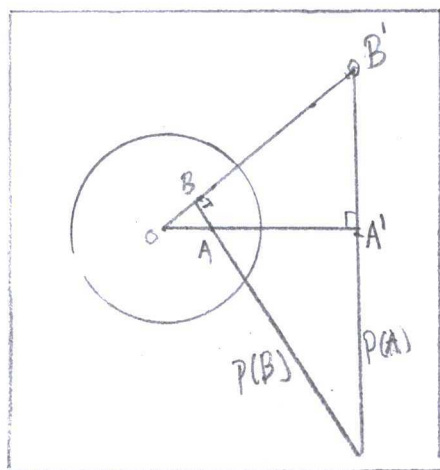
Proposition 4.3 (Midpoints)

Proposition 2.5

(pg 359) "Definition-Radical axis"

(pg 36) "Definition-Non-intersecting coaxal pencils"

P-12



(From P-exercise) "B lies on $p(A)$ if and only if A lies on $p(B)$."

"Polar reciprocity for a circle"

Proof: Hypothesis

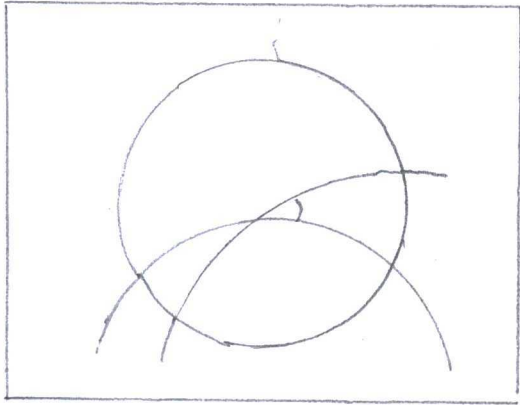
Evidence

K-12.

1) Circle γ has center O	Euclid's Postulate III
2) Point A is inside circle γ	(pg 130) "Definition - inside"
3) A' is an inverse to A	Proposition 7.5
4) Point B is also inside γ	(pg 130) "Definition - inside"
5) B' is the inverse to B	Proposition 7.5
6) Point P is the intersection between $\overline{B'A}$ and $\overline{BA'}$	Proposition 2.1
7) $\triangle OAB'$ is similar to $\triangle OBA'$ because $r^2 = (\overline{OA})(\overline{OA'})$ $= (\overline{OB})(\overline{OB'})$ $\frac{(\overline{OA'})}{(\overline{OB})} = \frac{(\overline{OB'})}{(\overline{OA})} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$	Corollary to Proposition 6.2 "Power of O " "Similar sides"
8) $\angle OBA' = \angle OA'B$ $= 90^\circ$	Euclid's Postulate IV
9) $\angle OBA' = \angle A'AP$	Axiom C-1
10) $\angle OBP = \angle OBA$	Axiom C-1

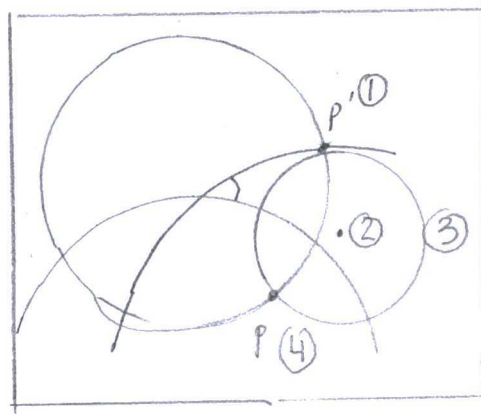
$$\begin{aligned} 11) \Delta OA'B &\propto \Delta PBB' \\ &\propto \Delta OAB' \end{aligned}$$

P-13



"Acute angle in the Poincaré model"

(from P-exercise) "Given an acute angle in the Poincaré model. Construct a Poincaré line that is perpendicular to a given side of the angle and limiting parallel to the other."

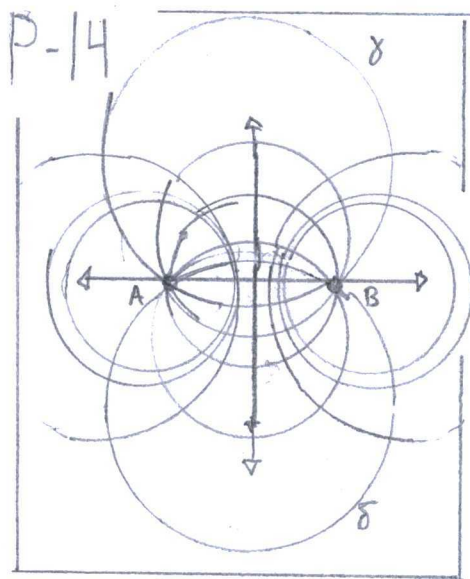


① Point P' for inversion

② The orthogonal circle center

③ An orthogonal circle with radius OP'

④ The point P .



a) (from P-exercise) "The set of all circles orthogonal to two given circles γ and δ tangent coaxal pencil through C whose line of centers is the common tangent $\vec{t}$ to γ and δ "

"Orthogonal
coaxal pencils

Proof: Hypothesis

Evidence

- 1) Circles γ and δ intersect at one point C and D
- 2) Line $\vec{t}$ is perpendicular to both γ and δ through C

Circle-Circle
Continuity Principle

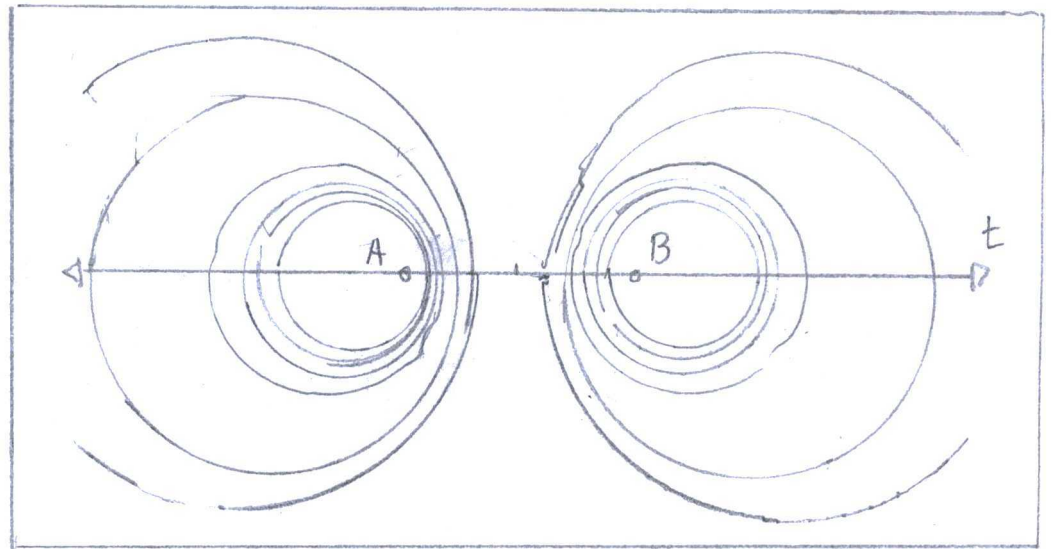
Proposition 2.5

- 3) γ, δ , and $\vec{t}$ are a tangent coaxal pencil

(pg 361) "Definition-
tangent coaxal
pencil.

b) (from P-exercise) "The set of circles orthogonal to two given non-intersecting non-concentric circles γ

Coaxal pencil whose line of centers is AB and whose radical axis is the perpendicular bisector of AB ."



"Non-intersecting non-concentric coaxal Pencil"

Proof: <u>Hypothesis</u>	Evidence
1) Circles γ and δ intersect at no points with centers A and B .	Euclid's Postulate III
2) A line $\overleftrightarrow{t}$ passes on AB .	Axiom I-1
3) Midpoint of AB has a radical axis	(pg 350) "Definition-radical axis"

4) The conditions are
Non-intersecting coaxial
pencils

(pg 361) "Definition-
Non-intersecting
coaxial pencil

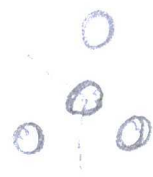
P-15

(From P-exercise) "Given three circles $\alpha, \beta,$ and γ . Is there always a fourth circle δ orthogonal to all three of them?

Yes! Three circles $\alpha, \beta,$ and γ have a radical axis, such as



a fourth orthogonal circle δ at the center between all the circles:



P-16.

a) (From P-exercise) "Inversion of δ maps the pencil of lines through P' onto the intersecting pencil of circles through O and P and maps the orthogonal pencil of concentric circles centered at P' onto the non-intersecting coaxial

pencil of circles whose radical axis
is the perpendicular bisector to $\overline{OP}$.

Proof: Hypothesis

Evidence

1) By the circle-circle continuity principle, σ and γ do meet in two points T and U	"Circle-Circle Continuity Principle"
2) Centers to σ and γ are O and Q	(pg 17) "Definition-center"
3) σ intersects γ at point O , and $\overline{OQ}$ at point P	Proposition 2.1
4) $\angle OTP$ and $\angle OUP$ are Scribe to the semi-circle of σ	Euclids Postulate III.31
5) $\overline{TU}$ crosses $\overline{OQ}$ at P'	Proposition 2.1
6) $\overline{PT}$ and $\overline{PU}$ are tangents to γ because rise ($\overline{P'T}$, $\overline{P'U}$) over run ($\overline{PP'}$).	(pg 202) "Definition-tangent"

7) P is the inverse of P'

8) Also P' is the inverse of P

9) The argument works for smaller circles

10) Further circles non-intersect on a coaxial of pencils

Proposition 7.2

RAA result

RAA hypothesis #2

RAA result and
(pg 361) "Definition-
Non-intersecting
Coaxial of
pencils"

b) (from P-exercise) "That inversion in γ maps the pencil of lines parallel to $\vec{l}$ onto the pencil of circles tangent to $\vec{l}$ at O "

Proof Hypothesis	Evidence
1) From the information de facto	RAA hypothesis
2) A line $\vec{l}$ through	Proposition 2.1

O.

3) $\vec{l}$ is tangent to δ

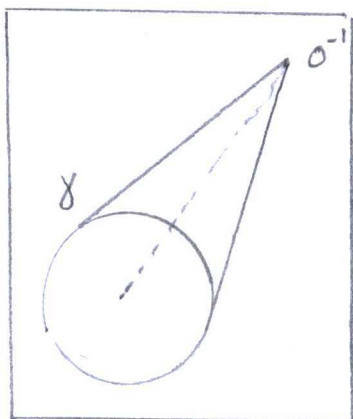
(pg 202) "Definition-Tangent"

4) $\vec{l}$ is tangent at each circle and pencil

RAA result

P-17.

a) (from P-exercise) "Inversion in a given "Circle" maps "circles" onto "circles."



Proof: Hypothesis

Evidence

1) "Circle" δ has a center O

Euclid's Postulate II

2) A point at ∞ is the inverse O' to O

Proposition 7.5

3) An inverse "circle" δ^{-1} to δ exists at ∞

Proposition 7.9.

b) (from P-exercise) "Circle δ' through A' and B' is orthogonal to α' by observing that inversion preserves orthogonality"

Proof: Hypothesis

Evidence

1) Points A and B
are inverse to
each other in
"circle" α

Proposition 7.9

2) Under inversion
in another "circle"
 β , A and B map
to A' and B',
ultimately α' .

Proposition 7.9

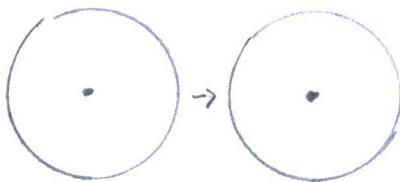
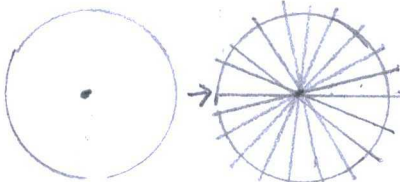
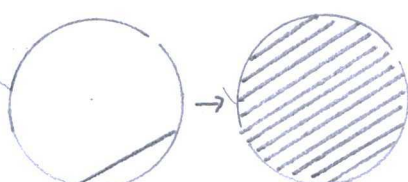
3) A' and B' are
inverses to each
other in α'

Proposition 7.5

4) Circle α' is
orthogonal to α

RAA result
by Proposition 7.9.

P-18.

		
"All circles have a given point as center"	"All lines passing through an ordinary point"	"A given line and all lines parallel to it"

a) (from P-exercise) "Two distinct "circles"
belong to a pencil of circles"

Proof: Hypothesis	Evidence
1) All of exercise P-II, exactly.	Reason being bedtime.

b) (from P-exercise) "The inversion of γ maps the pencil of lines parallel to l onto the pencil of circles tangent to l at O "

and δ is the intersecting coaxal pencil whose line of centers is the radical axis $\vec{E}$ of γ and δ and whose limiting points are the two points at which every member of this pencil cuts the line joining the centers of γ and δ

Proof: Hypothesis

Evidence

1) Circles δ and γ intersect at two points A and B.

Circle-Circle
Continuity Principle

2) Segment AB is line $\vec{E}$ in type II coaxal pencil.

(pg 361) "Definition-
Intersecting
coaxal pencil"

c) (from P-exercise) "The set of all circles orthogonal to two given circles γ and δ intersecting at A and B is the non-intersecting non-concentric

b) (from P-exercise) "A pencil of "circles"
is invariant as a set under
inversion in any "circle" in the pencil"

Proof Hypothesis	Evidence
<u>Case 1: All circles having a given point as center</u>	
A) Circle γ has radius OR	Euclids Postulate III
B) γ has a center, invariant to the pencil	(pg 365) "Definition- Circles in inversion plane"
<u>Case 2: All lines passing through a given ordinary point</u>	
A) Circle γ has a radius OR	Euclid's Postulate III
B) A point resides inside γ	(pg 310) "Definition- Ordinary point"
C) P-line crosses an	

ordinary point

(pg 302) "Definition-
P-line"

Case 3: A given line and
all lines parallel to it

A) Circle γ has radius
 $\overline{OR}$

Euclid's
Postulate III

B) P-line is in the
circle

(pg 302) "Definition-
P-line"

C) Parallel lines are
possible in Poincare
discs

(pg 304) "Definition-
Parallel in
Poincare disc"

P-19

(From P-exercise) " $\square ABCD$ is a regular 4-gon"

Proof: Hypothesis

Evidence

1) Points A, B, C, D reside
in a Poincare disc

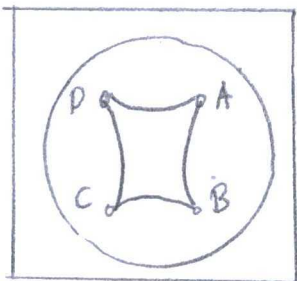
(pg 14) "Definition-
point"

2) Segments join points
from A to B, B to C,
C to D, and D to A.

(pg 14) "Definition-
segment"

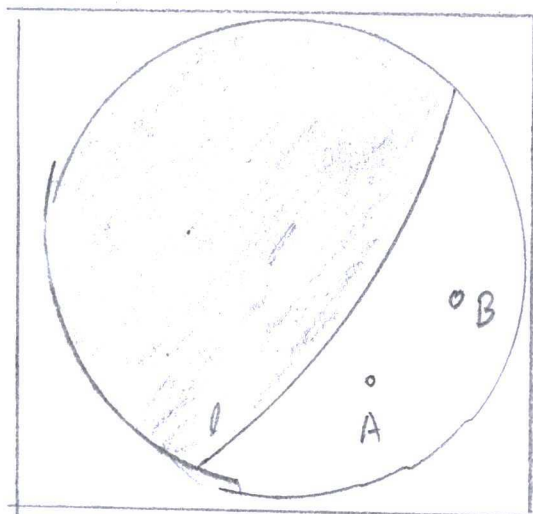
3) The shape is a quad-
rilateral 4-gon.

Proposition 7.12



"4-gon in the
Poincare disc"

P-20



"Same side"

(From P-exercise) "In the hyperbolic plane, there exists two points A, B lying on the same side of S of a line l such that no circle through A and B is entirely in S "

Proof: Hypothesis

Evidence

1) Two points A, B are in a Poincaré disc

(pg 14) "Definition-point"

2) A, B are on the same side of l

(pg 110) "Definition-Same side"

3) In the Euclidean plane, A, B cannot form a circle entirely in S because the Euclidean center is outside the Poincaré disc.

RAA hypothesis

4) A, B cannot form a P-circle

Proposition 7.12

5) The circle is not possible

RAA result